

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-012 - Nahorkatiya-012

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000134
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2022-11-10 / 2023-02-16
Measured / TVD	2329 m / 2309 m
Trajectory	Directional; inc 19.6 deg; az 326.6 deg
Formation Interval	1019-1678 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	2166 psi
Fracture Gradient	14.99 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (48.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	582	Surface
9 5/8 in casing	1350	Intermediate
7 in liner/casing	2049	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2022-12-26 1192 m	Torque Spike	Critical	Rotary speed fluctuated during constant pump rate. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2023-01-10 1533 m	Fishing Operation	High	Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Early stuck-pipe escalation may reduce probability of fishing.
2023-01-18 1707 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2022-12-19 1038 m	Stuck Pipe	Medium	High drag was recorded while circulating remained possible. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Avoid prolonged static periods in high-drag interval.
2022-12-26 1205 m	Stuck Pipe	Medium	High drag was recorded while circulating remained possible. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Avoid prolonged static periods in high-drag interval.
2023-01-11 1558 m	Torque Spike	Medium	Torque rose without corresponding pressure anomaly. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2022-12-04 695 m	NPT	Low	ROP dropped to zero during equipment unavailability. Mitigation: Verify equipment health before resuming drilling. Lesson: Recurring equipment downtime should influence planning and spares.

Completion Summary

The synthetic well reached 2329 m MD in the Tipam interval. The NWIS knowledge record links 8 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.